Dept. of Computer and Communication Engineering

Faculty of Computer Science and Engineering

Patuakhali Science and Technology University

Dumki, Patuakhali-8602, Bangladesh

Final Examination of B. Sc. Engineering in CSE Level: 3 Semester: I Session: 2017-2018

Course Code	**Course Title**	**January-June** 2020	**Credit:** 03
CCE 311	Numerical Methods		**Time:** 03 Hr
			Marks: 70

Answer any 05 out of 06 Questions (Split answers are highly discouraged)

1 [A.] Resolve the following system using the Cramer Rule. **7**

$$0.40X_1-0.1X_2-0.2X_3=7.85$$
$$0.10X_1+7X_2-0.3X_3=-19.3$$
$$0.30X_1-0.2X_2+10X_3=71.4$$

[B.] Use the Gauss-Elimination technique to resolve the following system. **7**

$$0.3X_1-0.1X_2-0.2X_3=7.85$$
$$0.10X_1+7X_2-0.3X_3=-19.3$$
$$X_1-0.2X_2+10X_3=71.4$$

2 [A.] Apply the Choleski's process to locate the root of the following system **7**

$$X_1+X_2-X_3=2$$
$$2X_1+3X_2+5X_3=-3$$
$$3X_1+2X_2-3X_3=6$$

[B.] **7**

i. How can we eliminate the error in the Trapezoidal rule by applying the Simpson's rule?

ii. Integrate using Simpson's 3/8 rule.

$f(x)=0.2+25x-200x^2+675x^3-900x^4+400x^5$ from a=0 to b=0.8

3 [A.] Solve the following system using the Gauss-Jordan method. **7**

$$0.7X_1-0.1X_2-0.2X_3=7.85$$
$$0.10X_1+7X_2-0.3X_3=-19.3$$
$$X_1-0.2X_2+10X_3=71.4$$

[B.] **7**

i. Show two scenarios in the case of an iteration process, which are convergence and divergence.

ii. Write down the algorithm of Bisection Method.

4 [A.] How to fit a second-order polynomial on the data of the following table. Also find out standard error $S_{y/x}$ from this table data. **7**

x_i	y_i
0	2.1
1	7.7
2	13.6

[B.] What is linear regression? Derive an equation for linear regression. How to estimate the parameter a_0 and a_1 through linear regression? **7**

5 [A.] **7**

i. State Weddle's Rule.

ii. Show that $x_r=x_u-f(x_u)(x_l-x_u)/(f(x_l)-f(x_u))$ in case of false position method.

[B.] Given dy/dx= ½ (x+y), y(0)=2, y(0.5)=2.636, y(1.0)=3.595, y(1.5)=4.968. Find y(2) by Milne's method. **7**

6 [A.] Given dy/dx=1+xy and y(0)=1. Calculate y(0.1), y(0.2) using Picard's method. **7**

[B.] Use the Euler's method to numerically integrate $dy/dx = -2X^3 + 12X^2 - 20X + 8.5$ from X=0 to X=4.0 with a step size 0.5. The initial condition at X=0 is Y=1. **7**

Patuakhali Science and Technology University

Final Examination of B.Sc. Engineering in CSE, Jan-June 2019

Course Code: CCE 311 **Course Title:** Numerical Methods **Credit:** 03

Time: 03 Hr

Answer any 05 out of 06 Questions **(Split answers are highly discouraged)** **Marks: 70**

1 A Use the Gauss-Jordan technique to solve the system. 7

$$3x_1 - 0.1x_2 - 0.2x_3 = 7.85$$
$$0.1x_1 + 7x_2 - 0.3x_3 = -19.3$$
$$0.3x_1 - 0.2x_2 + 10x_3 = 71.4$$

1 B Derive equation for linear regression and find out a_0 and a_1. 7

2 A Apply factorization process to the symmetric matrix. 7

$$[A] = \begin{bmatrix} 6 & 15 & 35 \\ 15 & 55 & 225 \\ 55 & 225 & 979 \end{bmatrix}$$

2 B Given $dy/dx = \frac{1}{2}(x+y)$, $y(0)=2$, $y(0.5)=2.636$, $y(1.0)=3.595$, $y(1.5)=4.968$. Find y(2) by Milne's method. 7

3 A Use the Euler-Cauchy's method to numerically integrate $dy/dx = -2X^3 + 12X^2 - 20X + 8.5$ from X=0 to X=3.0 with a step size 0.5. The initial condition at X=0 is Y=1. 7

3 B 7

i. Use Simpson's rule to integrate $f(x)=0.2+25x-200x^2+675x^3-900x^4+400x^5$ from $a=0$ to $b=0.8$

ii. Use it in conjunction with Simpson's 1/3 rule to integrate the same function for five segments.

4 A Use bisection method to solve the following problem up to approximate percent relative error $\varepsilon_a <= 0.422$. 7

$$f(c) = \frac{667.38}{c}(1 - e^{-0.146843c}) - 40$$

4 B Given $dy/dx = 1 + xy$ and y(0)=1. Calculate y(0.4), y(0.2) using Picard's method. 7

5 A 7

i. Drive the formula of false position method with proper figure.

ii. Differentiate between Gauss-Jordan and Gauss elimination method.

5 B Fit a second-order polynomial to the data of the following table. Also find out standard error $S_{y/x}$ 7

x_i	y_i
1	7.7
2	13.6
3	27.2

6 A Show a case where bisection is preferable to false position with a suitable example. 7

6 B Clear the concepts of convergence and divergence using iteration method with a suitable example. 7

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Faculty of Computer Science and Engineering
Patuakhali Science and Technology University

Final Examination of B.Sc. Engineering in CSE Level: 3 Semester: I Session: 2015-16

Course Code	Course Title	January-June	Credit: 03
CCE 311	Numerical Methods	2018	Time: 03 Hr
			Marks: 70

Answer any 05 out of 06 Questions (**Split answers are highly discouraged**)

Put appropriate figure/example if necessary

1 A Determine the real roots of f(x) = -0.4 x^2 + 2.2 x + 4.7 — 2+4

- I. Using the quadratic formula
- II. Using three iterations of the bisection method to determine the highest root. Employ initial guesses of lower $x_l = 5$ and $x_u = 10$. Compute the estimated error ε_a and true error ε_t after each iteration.

1 B Explain accuracy and precision with example. Write down different types of error found in numerical analysis with the formula to calculate them. — 2+3

1 C Derive the formula of false position method with proper figure. — 3

2 A How do you use Taylor series in the derivation of Newton-Raphson method? Explain the pitfalls of Netwan-Raphson method with example. — 3+3

2 B Use simple fixed-point iteration to locate the root of f(x) = -0.9 x^2 + 1.7x + 2.5 Use an initial guess of $x_0 = 5$ and iterate until $\varepsilon_a <= 0.01\%$ — 5

2 C State the differences between secant and false-position method. — 3

3 A Use Mullar's method with guesses of x0, x1and x2 = 4.5, 3.5 and 5.5 respectively, to determine a root of the equation — 6
f(x) = x^3 -13x – 12
Note that the roots of this equation are -3, -1, and 4

3 B Apply Cholesky decomposition to the symmetric matrix — 5

$$[A] = \begin{bmatrix} 6 & 15 & 35 \\ 15 & 55 & 225 \\ 55 & 225 & 979 \end{bmatrix} \quad \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad \begin{bmatrix} L_{11} & 0 & 0 \\ L_{21} & L_{22} & 0 \\ L_{31} & L_{32} & L_{33} \end{bmatrix}$$

3 C Write short notes on — 1.5+1.5

- i. LU decomposition
- ii. Lower and Upper triangular matrix

4 A Fit a second-order polynomial to the data of the following table. Also find out standard error $S_{y/x}$ — 7

x_i	y_i
1	7.7
2	13.6
3	27.2

4 B — 7

- i. Use Simpson's 3/8 rule to integrate $f(x)=0.2+25x-200x^2+675x^3-900x^4+400x^5$ from a=0 to b=0.8
- ii. Use it in conjunction with Simpson's 1/3 rule to integrate the same function for five segments
- iii. Find out E_t for both cases. true val = 1.640533

5 A Derive equation for linear regression and show criteria for a "Best" fit. — 7

5 B Given dy/dx= ½ (x+y), y(0)=2, y(0.5)=2.636, y(1.0)=3.595, y(1.5)=4.968. Find y(2) by Milne's method. — 7

6 A Use the point-slope method to numerically integrate $dy/dx = -2X^3 + 12X^2 - 20X + 8.5$ from X=0 to X=4.0 with a step size 0.5. The initial condition at X=0 is Y=1. — 7

6 B Given dy/dx=1+xy and y(0)=1. Calculate y(0.1), y(0.2) using Picard's method. — 7

Patuakhali Science and Technology University

5^{th} semester (L-3, S-I) Final Examination of B.Sc. in Engg. (CSE), Jan-June-2016

Session: 2013-14, Course Code: CCE-311, Course Title: Numerical Methods

Marks-70, Time: 3 hours, Credit: 3.00

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]

Answer any 5 of the following questions.

1. a) Determine the normal equations for polynomial of the nth degree — 6

   b) Find the values of a, b and c so that $Y = a + bx + cx^2$ is best fit to the data — 4

x	y
0	1
1	0
2	3
3	10
4	21

   c) Fit a function of the form $y = ax^b$ to the following data — 4

x	y	x	y
2	43	20	8
4	25	40	5
7	18	60	3
10	13	80	2

2. a) Derive the LU decomposition method for solving a system of n linear equations — 7

   b) Solve the following system — 5

$$5x - 2y + z = 4$$
$$7x + y - 5z = 8$$
$$3x + 7y + 4z = 0$$

   By the unit lower triangular and upper triangular (LU) method.

   c) Write down the difference between Gauss elimination method and Gauss-Jordan method. — 2

3. a) Explain the term 'interpolation' — 3

   b) Deduce Gauss's interpolation formula — 6

   c) Using Gauss's forward formula, find the value of $f(32)$ given that $f(25)=0.2707$, $f(30)=0.3027$, $f(35)=0.3386$, $f(40)=0.3794$ — 5

4. a) Discuss the method of Bisection to find an approximate root of an equation $f(x) = 0$ — 6

   b) Find a real root of the equation $x^2 - x - 2 = 0$ by using Bisection Method and False position method. — 4+4=8

5. a) Describe the Taylor Series Method for the solution of ordinary differential equation. — 6

   b) Given $\frac{dy}{dx} = 1 + xy$ when $y(0) = 2$ find $y(0.1)$ by Taylor Series Method and Runge Kutta 2^{nd} order method. — 4+4=8

6. a) Derive general integration formula to compute $\int_a^b f(x)dx$ and Simpson's 1/3 rule for the numerical integration. — 3+4=7

   b) Compute the integral $\int_0^1 \frac{dx}{\sqrt{1+x^2}}$ upto 4 decimal place by using Trapezoidal rule and Simpson's 1/3. — 3+4=7

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Patuakhali Science and Technology University

5^{th} semester (L-3, S-I) Final Examination of B.Sc. in Engg. (CSE), Jan-June-2015

Session: 2012-13, Course Code: CCE-311, Course Title: Numerical Methods

Marks-70, Time: 3 hours, Credit: 3.00

[Figure in the right margin indicates full marks. Split answering of any question is not recommended.]

Answer any 5 of the following questions.

1. a) Define interpolation of a polynomial. 2

   b) Deduce Newton's interpolation formula for a polynomial. 5

   c) Values of x (in degrees) and $\sin x$ are given in the following table: 7

x (in degrees)	$\sin x$
15	0.2588190
20	0.3420201
25	0.4226183
30	0.50
35	0.5735764
40	0.6427876

Determine the values of $\sin 18^0$ and $\sin 38^0$

2. a) Determine the normal equations if the cubic polynomial $y = a_0 + a_1x + a_2x^2 + a_3x^3$ is fitted to the data points $(x_i, y_i), i = 1,2,\cdots,m$ 6

   b) Use the method of least squares to fit the straight line $y = a + bx$ to the data 4

x	y
0	2
1	5
2	8
3	11

Find the values of a and b.

   c) The curve $y = ce^{bx}$ is fitted to the data 4

x	y
1	1.5
2	4.6
3	13.9
4	40.1
5	125.1
6	299.5

Find the best values of c and b.

3. a) Derive the Gauss elimination method for a system of n linear equations and hence write the difference between Gauss method and Gauss-Jordan method. 7

b) Solve the following system 7

$$5x - 2y + z = 4$$
$$7x + y - 5z = 8$$
$$3x + 7y + 4z = 10$$

by (*i*) Gauss elimination method and (*ii*) Gauss-Jordan method.

4. a) Derive the Bisection Method and False position method. 8

b) Find a real root of the equation $(x^3 + x^2 - 1 = 0$ by using False position Method. 6

5. a) Derive the Trapezoidal rule and simpson's 1/3 rule. 7

b) Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by using Trapezoidal rule and Simpson's 1/3. 7

6. a) Describe the Runge Kutta 2nd order method for the solution of first order ordinary differential equation. 7

b) Solve $\frac{dy}{dx} = x + y$, $y(0) = 1$ and find $y(0.2)$ by Runge Kutta 4th order method. 7

Patuakhali Science and Technology University

B.Sc. Engg. (CSE) Level: 3 Semester: I Final Examination of Session 2011-12, Jan-June 2014

Course Code: CCE 311 Course Title: Numerical Methods

Credit Hour: 03 Full Marks: 70 Duration: 03 Hours

[Figures in the right margin indicate full marks. Split answering of any question is highly discouraged. Write the full question number e.g. 1(B)(i) before the answer paragraph]

Answer any 5 of the following questions

1 A Use Gauss-Elimination technique to solve the following system 7

$$0.2x_1-0.7x_2-0.2x_3=7.85$$
$$0.10x_1+7x_2-0.3x_3=-19.3$$
$$0.30x_1-0.2x_2+x_3=71.4$$

1 B i. How can we remove the error of Trapezoidal rule by applying Simpson's rule? 7

ii. Sketch and state graphical depiction of singular and ill-conditioned systems.

2 A Show a case where Bisection is preferable to False Position. 7

2 B i. State Weddle's Rule.

ii. Solve the following system using Cramer's rule 7

$$0.3x_1-0.1x_2-0.2x_3=7.85$$
$$0.7x_1+0.7x_2-0.3x_3=-19.3$$
$$0.30x_1-0.2x_2+10x_3=71.4$$

3 A Apply Factorization process to find out the root of the following system 7

$$x_1+x_2-x_3=2$$
$$2x_1+3x_2+5x_3=-3$$
$$3x_1+2x_2-3x_3=6$$

3 B i. Write down the algorithm of Bolzano's Method.

ii. Use Simpson's 3/8 rule to integrate 7

$$f(x)=0.2+25x-200x^2+675x^3-900x^4+400x^5 \text{ from } a=0 \text{ to } b=0.8$$

4 A Show the effect of reduced step size on Euler's method. 7

B i. Show two cases of iteration process. 7

ii. What is Lagrange's interpolation formula? In which case we use it?

5 A "Secant method can be classified as a bracketing method" --- is it true or false? Explain your answer clearly. 7

5 B i. Develop equation for linear regression. 7

ii. Show that $x_r=x_u-f(x_u)(x_l-x_u)/(f(x_l)-f(x_u))$ in case of false position method.

6 A Use N-R method to estimate the root of $f(x)=e^{-x}-x$, employing an initial guess of $x_0=0$ (True value=0.56714329) until $\varepsilon_t(\%)<10^{-8}$. 7

6 B Use Gauss-Jordan technique to solve the following system 7

$$3x_1-x_2-0.2x_3=7.85$$
$$0.10x_1+7x_2-0.3x_3=-19.3$$
$$0.30x_1-0.2x_2+x_3=71.4$$

2014-15